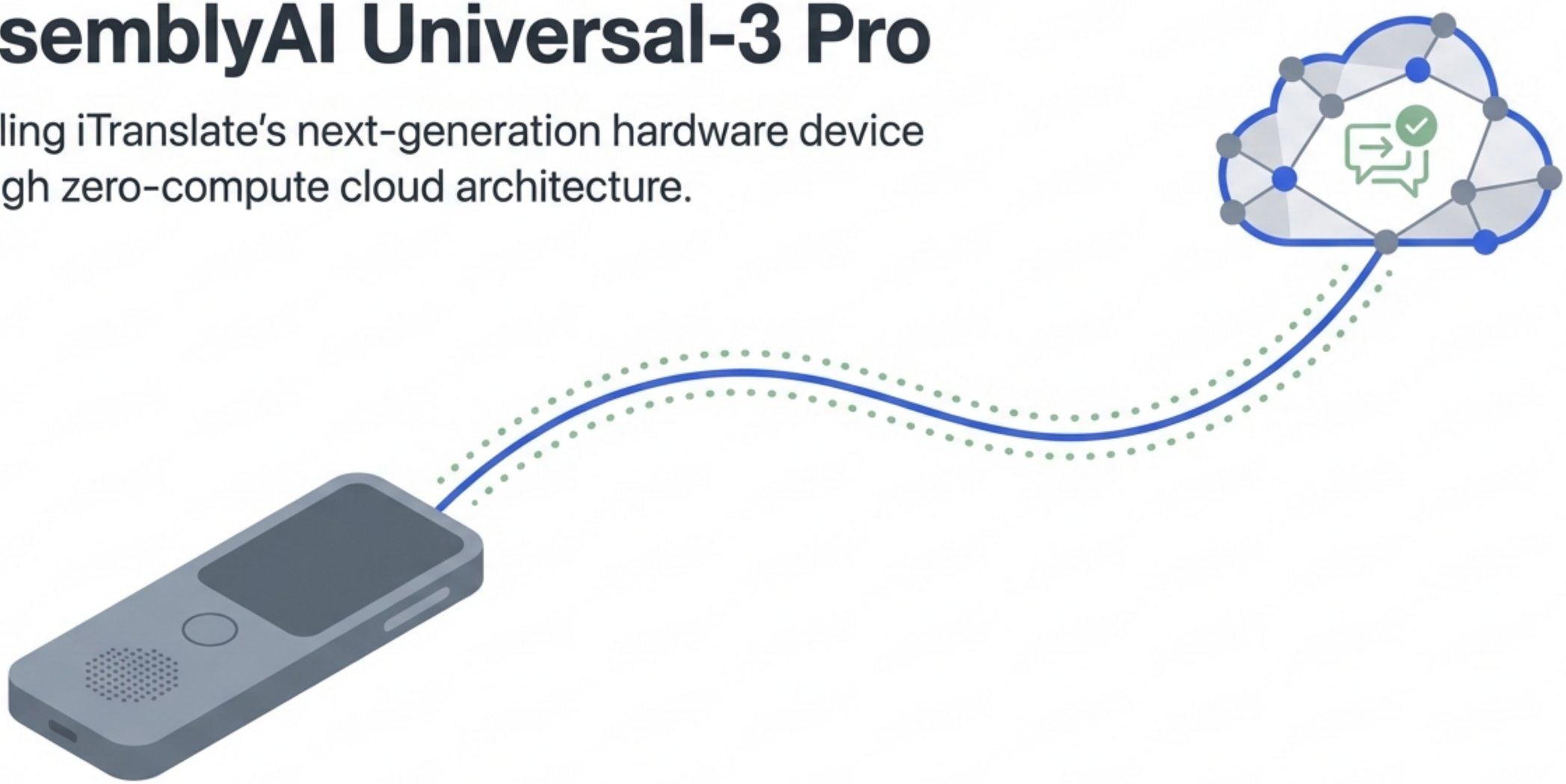
Real-Time Translation with AssemblyAI Universal-3 Pro

Enabling iTranslate's next-generation hardware device through zero-compute cloud architecture.



Edge computing bottlenecks throttle real-time translation hardware

The Device

Portable, battery-powered handheld translator strictly constrained by a lack of a dedicated GPU.

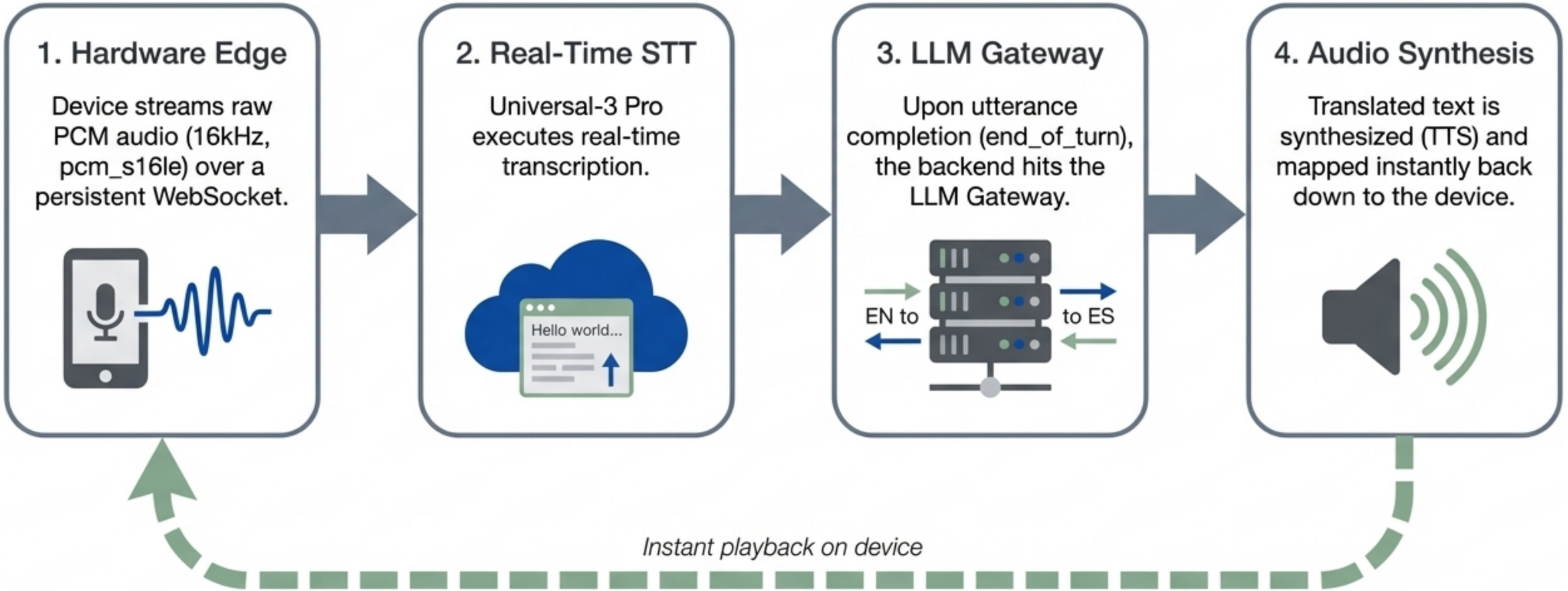
The Goal

Flawless, real-time conversational performance, launching with English ↔ Spanish and expanding to 10 language pairs.

The Constraint

Insufficient on-device compute for advanced ML model inference, highly accurate external STT feed that does not inflate the hardware Bill of Materials (BOM).

Cloud-centric architectures shift the heavy ML lifting off the device



Universal-3 Pro Streaming is purpose-built for low-latency conversational AI

Code-Switching

Native multilingual support effortlessly handles English/Spanish mixing naturally within a single utterance without manual language swapping.

Keyterms Prompting

Biases recognition for brand names and medical terms at live inference and turn-level post-processing via the keyterms_prompt parameter.

Voice Agent Optimized

Engineered specifically to clear the 300ms latency rule applied to short conversational utterances.

Zero Device Compute

Eliminates heavy on-device processing, creating massive BOM cost savings and enabling state-of-the-art models without firmware updates.

Native SDKs eliminate complex WebSocket infrastructure

Python Backend
Orchestration Layer

Serves as the Orchestration Layer, enabling the team to focus strictly on translation business logic rather than managing raw audio streams.

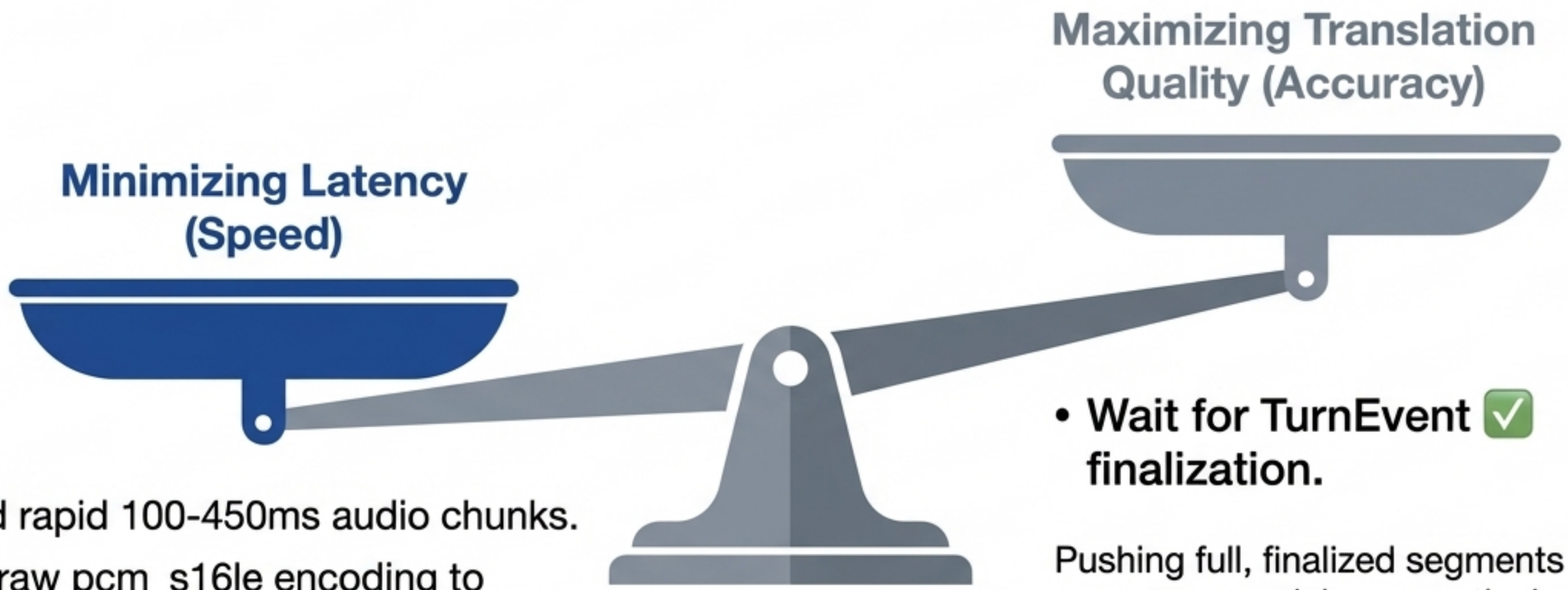
AssemblyAI Streaming API

TypeScript
Control Plane

Operates as the Control Plane, managing the companion apps and UI status feeds perfectly in sync with the transcription state.

Developers integrate natively in hours, bypassing raw infrastructure maintenance.

Granular latency controls optimize the 4G LTE handheld experience

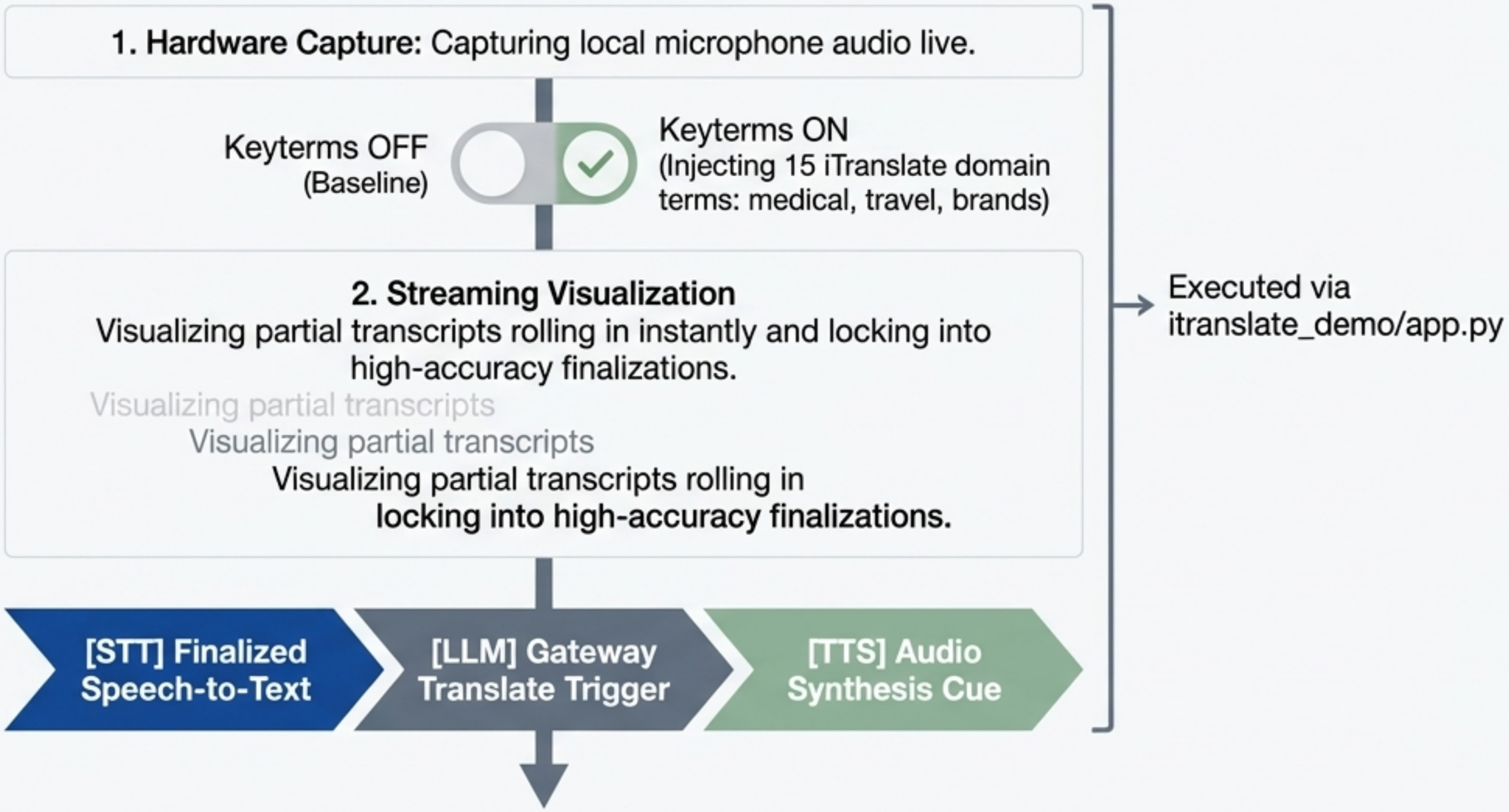


- Send rapid 100-450ms audio chunks.
- Use raw pcm_s16le encoding to eliminate transcoding payload bloat.
- Disable native formatting if the LLM resolves punctuation natively.

- **Wait for TurnEvent ✅ finalization.**

Pushing full, finalized segments preserves crucial grammatical alignment and context for the LLM compared to disjointed word-by-word streaming.

Tracing the translation pipeline over a single persistent session



Cloud deployment unlocks infinite scale without firmware updates

Current State: Hardware Constraints

Future-Proofing

Seamlessly add 10+ language pairs immediately without impacting device storage limits or forcing user firmware updates.

DevOps Elimination

The backend effortlessly auto-scales to handle thousands of concurrent device sessions, completely eliminating backend infrastructure management for the iTranslate team.

Ephemeral audio streaming guarantees enterprise-grade privacy

Zero Audio Retention

Streaming endpoints evaluate audio strictly ephemerally. Audio is permanently discarded immediately after the session concludes, securing sensitive user translations.

Data Privacy

End-to-end data encryption in transit via WSS and strictly encrypted at rest.

Enterprise Security

AssemblyAI maintains rigorous compliance and is fully SOC 2 Type 2 Certified.

Foundation of Trust

Accelerate time-to-market with a joint technical pilot

01

Phase 1: Technical Kickoff

Schedule a deep-dive architecture session connecting iTranslate engineers with AssemblyAI experts.

02

Phase 2: Device Pilot

Deploy a pilot program pushing AssemblyAI directly to the iTranslate test device fleet for real-world validation.

03

Phase 3: Engineering Support

Leverage direct access to AssemblyAI's Applied AI Engineering team to bypass roadblocks and accelerate the launch timeline.

Ready to Start? Schedule your architectural review.